

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A \end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

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Differentiation***Pembezaan***

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule***Petua Hasil Tambah***

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule***Petua Hasil Darab***

$$\frac{d}{dx}(u(x) \cdot v(x)) = v \frac{du}{dx} + u \frac{dv}{dx}$$

Quotient Rule***Petua Hasil Bahagi***

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain Rule***Petua Rantai***

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$$

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Shape <i>Bentuk</i>	Volume <i>Isipadu</i>	Surface Area <i>Luas Permukaan</i>
Sphere <i>Sfera</i>	$V = \frac{4}{3} \pi r^3$	$S = 4 \pi r^2$
Right circular cone <i>Kon membulat tegak</i>	$V = \frac{1}{3} \pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder <i>Silinder membulat tegak</i>	$V = \pi r^2 h$	$S = 2 \pi r^2 + 2 \pi r h$

SECTION A [25 marks]

BAHAGIAN A [25 markah]

This section consists of 3 questions. Answer **all** questions.

*Bahagian ini mengandungi 3 soalan. Jawab **semua** soalan.*

1 Find
Cari

(a) $\lim_{x \rightarrow 3} \frac{\sqrt{x^2+7} - \sqrt{2x+10}}{x-3}$

[5 marks]

[5 markah]

(b) Given the function
Diberi fungsi

$$f(x) = \begin{cases} \frac{x^2 - a^2}{x - a}, & x < a \\ x + 2, & x \geq a \end{cases}$$

Find the value of a , such that $f(x)$ is continuous at $x = a$

[4 marks]

[4 markah]

2 Given
Diberi

$$5e^{3y} = x^{-2} \sin^2 x$$

Show that

Tunjukkan bahawa

$$\frac{dy}{dx} = a \left(\frac{1}{x} \right) + b \cot x$$

where a and b are constants. State the value of a and b .

dimana a dan b adalah pemalar. Nyatakan nilai a dan b .

[9 marks]

[9 markah]

3 Given
Diberi

$$f(x) = a \ln x + bx + \frac{6}{x}, x > 0$$

and its stationary points are
dan titik pegunnya ialah

$$x = 1 \text{ and } x = -3$$

Find the value of constant a and b .

Cari nilai pemalar a dan b

[7 marks]

[7 markah]

SECTION B [75 marks]

BAHAGIAN B [75 markah]

This section consists of 7 questions. Answer **all** questions.

*Bahagian ini mengandungi 7 soalan. Jawab **semua** soalan.*

1 Given

Diberi

$$z = -\sqrt{3} - i$$

State z in polar form.

Nyatakan z dalam bentuk kutub

[5 marks]

[5 markah]

2 (a) Solve the equation

Selesaikan persamaan

$$\ln 2^{4x-1} = \ln 8^{x+5} + \log_2 16^{1-2x}$$

by expressing your answer in terms of $\ln 2$

dengan mengungkapkan jawapan anda dalam sebutan $\ln 2$

[4 marks]

[4 markah]

(b) Determine the solution set for the following inequalities

Tentukan set penyelesaian bagi ketaksamaan berikut

i) $3(x + 2) < 7 - 2x$

ii) $\frac{x^2 - 3x - 4}{x + 1} \leq 2$

[5 marks]

[5 markah]

Hence find the solution set for both $3(x + 2) < 7 - 2x$ and $\frac{x^2 - 3x - 4}{x + 1} \leq 2$.

Seterusnya cari penyelesaian bagi kedua-dua $3(x + 2) < 7 - 2x$ dan $\frac{x^2 - 3x - 4}{x + 1} \leq 2$

[2 marks]

[2 markah]

- 3 Given the function $f(x) = e^{ax} + b$ and $(g \circ f)(x) = \frac{1}{2} \ln(e^{ax} + c)$

Diberi fungsi $f(x) = e^{ax} + b$ dan $(g \circ f)(x) = \frac{1}{2} \ln(e^{ax} + c)$

- (a) Express $g(x)$ in terms of b and c . State the domain of $g(x)$ in terms of b and c
Ungkapkan $g(x)$ dalam sebutan b dan c . Nyatakan domain $g(x)$ dalam sebutan b dan c

[7 marks]

[7 markah]

- (b) Hence find $g^{-1}(x)$ in terms of b and c
Seterusnya cari $g^{-1}(x)$ dalam sebutan b dan c

[3 marks]

[3 markah]

- (c) Given that the value of $b = 3$ and $c = 2$. Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes.

Diberi nilai $b = 3$ dan $c = 2$. Lakarkan graf bagi $g(x)$ and $g^{-1}(x)$ pada paksi yang sama.

[3 marks]

[3 markah]

- 4 Given the function $f(x) = \sqrt{5x + 4}$ and $g(x) = \frac{2x-3}{x+1}; x \neq -1$. Find

$(f \circ g^{-1})(0)$

Diberi fungsi $f(x) = \sqrt{5x + 4}$ dan $g(x) = \frac{2x-3}{x+1}; x \neq -1$. Cari

$(f \circ g^{-1})(0)$

[7 marks]

[7 markah]

- 5 Given
Diberi

$$f(x) = \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)}$$

- (a) State the domain of $f(x)$.

Nyatakan domain bagi $f(x)$.

[1 marks]

[1 markah]

- (b) Find the x and y intercept of $f(x)$.

Cari pintasan x dan y bagi $f(x)$.

[4 marks]

[4 markah]

- (c) Find the vertical and horizontal asymptotes of $f(x)$.

Cari asimptot mencancang bagi $f(x)$.

[6 marks]

[6 markah]

- (d) Hence, sketch the graph of $f(x)$.

Seterusnya, lakar graf $f(x)$.

[3 marks]

[3 markah]

- 6 Given the curve equation $y = x + \ln(\cos y)$
Diberi persamaan lengkung $y = x + \ln(\cos y)$

- (a) Find $\frac{dy}{dx}$. Hence show that $\frac{d^2y}{dx^2} + \frac{\sec^2 y}{(1 + \tan y)^3} = 0$

Cari $\frac{dy}{dx}$. Seterunya tunjukkan bahawa $\frac{d^2y}{dx^2} + \frac{\sec^2 y}{(1 + \tan y)^3} = 0$

[8 marks]

[8 markah]

- (b) A curve is defined by the parametric equations

Satu lengkung ditakrifkan oleh persamaan parametrik berikut

$$x = 2t^2 - 3t \qquad y = \frac{t^2 + 2t}{t-1}$$

(i) Find $\frac{dy}{dx}$

Cari $\frac{dy}{dx}$

Hence, find the equation of the tangent to the curve at $t = 2$

Seterusnya, cari persamaan tangen kepada lengkung pada titik $t = 2$

[8 marks]

[8 markah]

- 7 Find positive real numbers x and y which maximise $P = x^2y$ subject to the constraint $2x + 3y = 60$. Hence find the maximum value of P

Cari nilai positif x dan y yang memaksimumkan $P = x^2y$ dengan syarat $2x + 3y = 60$. Seterusnya, tentukan nilai maksimum P .

[9 marks]

[9 markah]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT